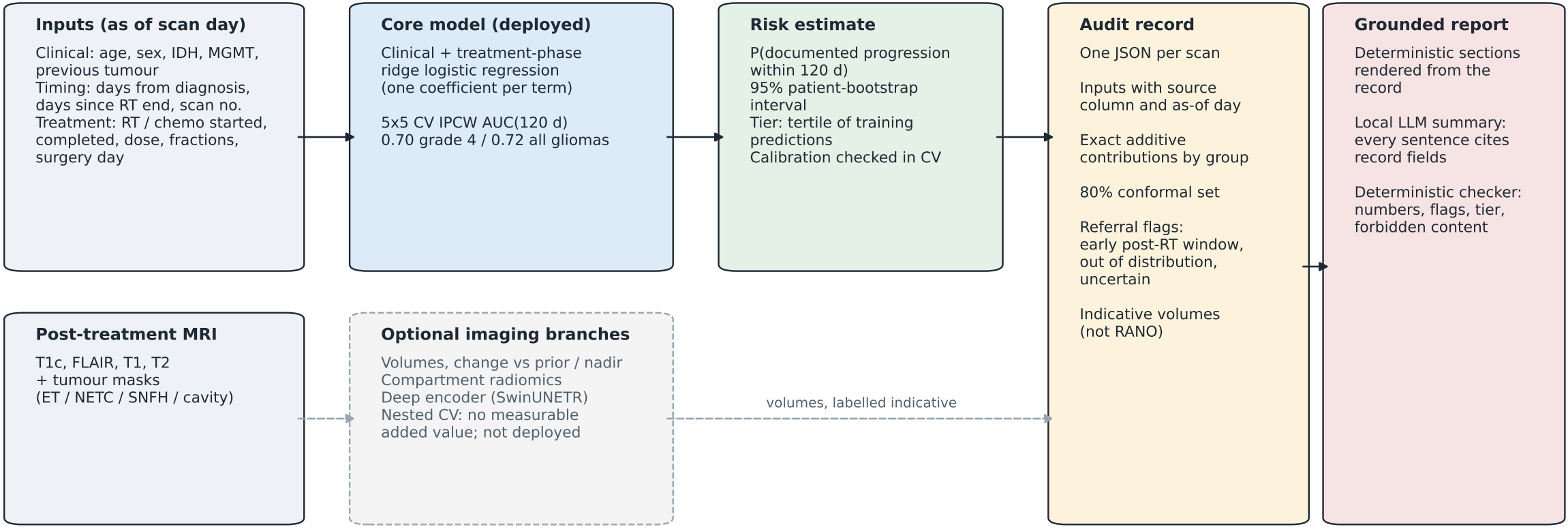
Inputs (as of scan day)
Clinical: age, sex, IDH, MGMT, previous tumour
Timing: days from diagnosis, days since RT end, scan no.
Treatment: RT / chemo started, completed, dose, fractions, surgery day
Core model (deployed)
Clinical + treatment-phase ridge logistic regression (one coefficient per term)
5x5 CV IPCW AUC(120 d)
0.70 grade 4 / 0.72 all gliomas
Risk estimate
P(documented progression within 120 d)
95% patient-bootstrap interval
Tier: tertile of training predictions
Calibration checked in CV
Audit record
One JSON per scan
Inputs with source column and as-of day
Exact additive contributions by group
80% conformal set
Referral flags: early post-RT window, out of distribution, uncertain
Indicative volumes (not RANO)
Grounded report
Deterministic sections rendered from the record
Local LLM summary: every sentence cites record fields
Deterministic checker: numbers, flags, tier, forbidden content
Post-treatment MRI
T1c, FLAIR, T1, T2
+ tumour masks
(ET / NETC / SNFH / cavity)
Optional imaging branches
Volumes, change vs prior / nadir
Compartment radiomics
Deep encoder (SwinUNETR)
Nested CV: no measurable added value; not deployed
volumes, labelled indicative